Identification of QTLs Impacting Kernel Discoloration Resistance and Grain Protein Content in Barley (*Hordeum vulgare* L.)

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Introduction

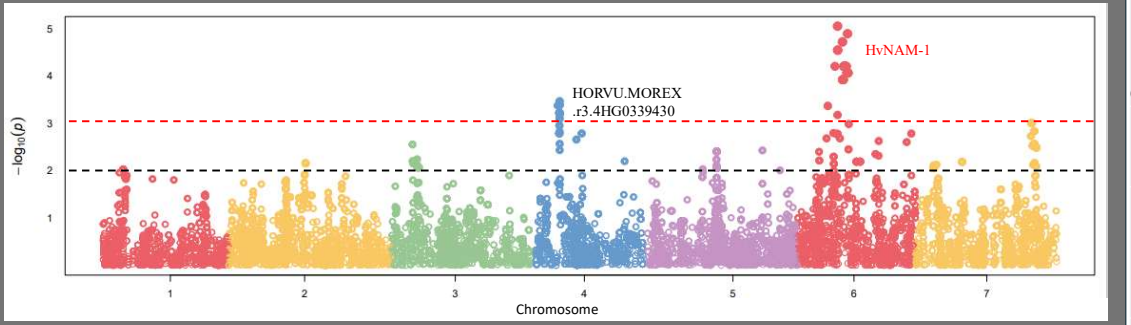
- Kernel discoloration is used by maltsters as a visual assay to avoid purchasing low-quality grain.
- Discoloration is influenced by many factors but is co-located with the low-protein gene on chromosome 6H from the cultivar 'Karl'.
- The low-protein gene from 'Karl' has been used by many programs to reduce the grain protein content to acceptable levels (i.e., <13.0%).

Materials and Methods

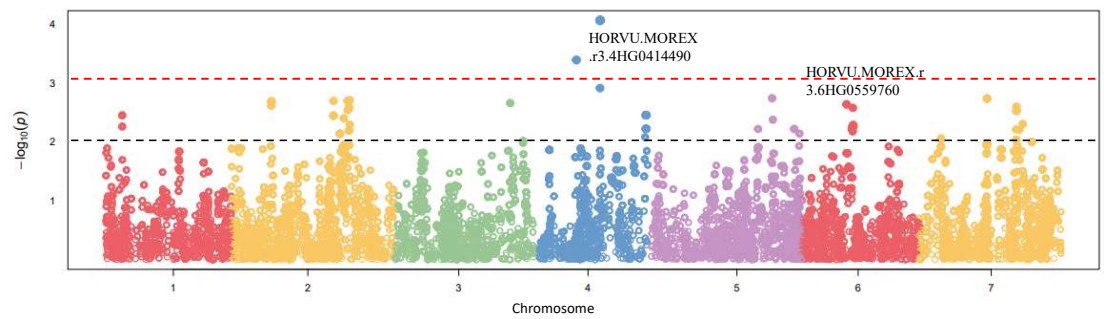
•168 lines from NDSU breeding program were evaluated at three locations in North Dakota (2023).
•Kernel color measured via spectrophotometer (L-value).
•Grain protein content assessed via near-infrared reflectance.
•Best linear unbiased estimates (BLUPs) generated for each line across locations using mixed model equation.
•Genotyping: 50K Illumina Infinium iSelect array.
•SNP data filtered (MAF ≤ 0.05, missing data ≥ 0.10).
•GWAS performed to identify significant QTLs using the BLUPs.

Spring malting barley lines with acceptable color and grain protein content could be developed in the future.

Results from the GWAS analysis of grain protein content



Results from the GWAS analysis of kernel color (L-value)



Abstract and contact information

Results and Discussion

- Significant QTLs for kernel color and grain protein were identified on chromosomes 4H and 6H.
- Unsurprisingly, the significant QTLs on chromosome 6H were co-located in the centromeric region.
- The significant QTLs on chromosome 4H do not appear to be linked. The QTL for grain protein content is in the short arm, and the QTL for kernel color is in the long arm.

Chromosome	Region of Chromosome	Gene closest to most significant marker	Linked Trait
4H	Short Arm	HORVU.MOREX.r3.4HG0339430	Protein***
4H	Long Arm	HORVU.MOREX.r3.4HG0414490	Color***
6H	Centromeric	HvNAM-1 HORVU.MOREX.r3.6HG0559760	Protein*** & Color**

*, **, *** $P \leq 0.05$, $P \leq 0.01$, and $P \leq 0.001$, respectively; which is equivalent to $-log_{10}(P) \geq 1.3$, 2.0, and 3.0, respectively.

Future Directions

- Collect more grain protein and kernel color data (2024-2025) to determine if genetic haplotypes for each trait can be identified on chromosomes 4H and 6H.
- Test lines with the low-protein QTL on chromosome 4H and the high-protein QTL on chromosome 6H to if the lines have acceptable protein levels.

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